

CO1-ASSESSMENT TOOL-2

INTERNET PROGRAMMING CSA4302

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1: HTTP Response Analysis :

A web server returns the following HTTP response header:

HTTP/1.1 200 OK

Date: Mon, 01 Apr 2026 10:00:00 GMT

Content-Type: text/html Content-Length: -150

Connection: keep-alive

ANSWERS:

1. Invalid Field

Content-Length: -150 is invalid because the value cannot be negative. Content-Length should always be a positive number.

2. Content-Length Issue

A negative Content-Length confuses the browser while receiving data. The webpage may not load correctly.

3. Status Code

The status code **200 OK** means the request was successful. However, the invalid header makes the response incorrect.

4. Correction

Replace Content-Length: -150 with the correct file size. Keep all HTTP header values valid.

2: DOM Structure Analysis:

Consider the following DOM tree representation:

```
<html>
<head>
<title>TestPage</title>
</head>
<body>
<div>
<p>HelloWorld
</div>
</body>
</html>
```

ANSWERS:

1. Missing Tags

The <p> tag is not closed properly. Every HTML tag should have the correct closing tag.

2. Browser Interpretation

Browsers try to fix missing tags automatically. Different browsers may display the page differently.

3. Rendering

The webpage may show incorrect formatting because of the invalid HTML structure. Proper tags avoid layout issues.

4. Correction

Close all HTML tags correctly and follow HTML5 syntax. This makes the DOM valid.

3: GET vs POST in Login System :

A login system uses two different methods:

Method	URL Example	Data Visibility	Security Level
GET	/login?user=admin&pass=1234	Visible in URL	Low
POST	/login	Hidden in body	Moderate

ANSWERS:

1. Difference

GET sends data in the URL, while POST sends data in the request body. POST is safer for sensitive information.

2. Suitable Method

POST is better for login forms because passwords are not shown in the URL. It provides better security.

3. GET Risk

Passwords sent through GET are visible in the browser history and URL. This creates a security risk.

4. Best Practice

Use POST with HTTPS for secure login. Always validate user input on the server.

4: HTML Code Inspection:

Analyze the following HTML snippet:

```
<!DOCTYPE html>

<html>

<head>

<title>Sample</title>

</head>

<body>

<h1>Welcome

<p>This is a paragraph



</body>

</html>
```

ANSWERS:

1. Missing Closing Tags

The <h1> and <p> tags are not closed. Closing tags are required for proper HTML structure.

2. Image Tag

The tag should include an alt attribute. It improves accessibility and displays alternate text if the image cannot load.

3. Browser Handling

Browsers may automatically correct some HTML errors. However, the page may not appear the same in every browser.

4. Correction

Close all missing tags and add the alt attribute to the image. This makes the HTML valid.

5: Browser-Server Communication Flow :

The following sequence describes a web request:

1. User enters a URL in the browser
2. DNS lookup occurs
3. HTTP request is sent
4. Server processes request
5. Response is returned
6. Browser renders page

ANSWERS:

1. Communication Flow

The browser finds the server using DNS, sends an HTTP request, and displays the response. This is the normal web communication process.

2. Delays

Delays can happen during DNS lookup, network transmission, or server processing. A slow internet connection can also increase loading time.

3. DNS Role

DNS converts the website name into an IP address. This helps the browser locate the correct server.

4. Optimization

Use fast DNS, caching, and optimized web pages. These methods reduce loading time and improve performance.